

Adjacent Vertices Fault-Tolerance Bifanability of Hypercube with the Same Color Sources *

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Abstract

Let $Q_n = (V_b \cup V_w, E)$ be the n -dimensional hypercube. Let F_a be the set of f_a pairs of adjacently faulty vertices. Let $s_1, t_1^1, \dots, t_1^{k_1}, s_2, t_2^1, \dots, t_2^{k_2} \in V_i$, $t_1^1, t_2^1 \in V_j$ be arbitrary fault-free vertices of Q_n for $\{i, j\} = \{b, w\}$. In this paper, we construct the spanning internally disjoint paths $P(s_1, t_1^i)$ and $P(s_2, t_2^j)$ of $Q_n - F_a$ for $f_a \leq n - 3$, $f_a + k_1 + k_2 = n - 1$, $1 \leq i \leq k_1$, $1 \leq j \leq k_2$.

1 Introduction

The hypercube network is one of the most popular and efficient topological structures of interconnection networks. This is partially due to its many excellent features, such as regularity, symmetry, low diameter and degree, high connectivity, simple routing algorithms, and so on. Component failures are unavoidable in a large parallel systems. Thus, fault tolerance is an important research issue for interconnection networks.

An interconnection network is usually represented by a graph where vertices represent processors and edges represent links between processors. Let $G = (V_b \cup V_w, E)$ be a bipartite graph where V_b and V_w are two disjoint vertex sets such that each edge of E consists of one vertex from each set. A bipartite graph $G = (V_b \cup V_w, E)$ is *Hamiltonian laceable* if there exists a *Hamiltonian path* between x, y for any $x \in V_b, y \in V_w$. The graph $G = (V_b \cup V_w, E)$ is *hyper-Hamiltonian laceable* if $\forall v \in V_b$ (resp. V_w), there exists a Hamiltonian path of $G - \{v\}$ between each pair of vertices of V_w (resp. V_b). In [7], Tsai et al. proved that

$Q_n - F_e$ is Hamiltonian laceable for $F_e \subset E(Q_n)$ and $|F_e| \leq n - 3$.

In [3], Chen et al. proposed some more general concepts. A k -fan $A(s \rightarrow T)$ of a bipartite graph $G = (V_b \cup V_w, E)$ is a set of k paths from $s \in V_b$ and $T = \{t_i | t_i \in V_w, \text{ and } t_j \in V_b, \text{ for } 2 \leq j \leq k\}$, such that every two paths of them share only the vertex s . Let $V(A(s \rightarrow T))$ be the set of vertices incident with some paths in $A(s \rightarrow T)$. A k^* -fan (s, T) of G is a k -fan with $V(A(s \rightarrow T)) = V(G)$. A graph G is k^* -fanable, if $\forall s \in V_i$ and $\forall T = \{t_1, t_2, \dots, t_k | t_1 \in V_j \text{ and } t_2, t_3, \dots, t_k \in V_i\}$ for $\{i, j\} = \{b, w\}$, there exists a k^* -fan. A graph G is f_e edges k^* -fanable if $G - F_e$ is k^* -fanable $\forall F_e \subset E(G)$ with $|F_e| = f_e$. Chen et al. proved that Q_n is f_e edges k^* -fanable for $n \geq 3, 0 \leq f_e \leq n - 2, 1 \leq k + f_e \leq n$ in [3].

In [4], the authors introduced the fanability with two source vertices with different color, named bifanability. A k -bifan with different color sources $B(s_1 \rightarrow T_1, s_2 \rightarrow T_2)$ of a bipartite graph $G = (V_b \cup V_w, E)$ is a set of k paths constructed by $|T_1|$ -fan $A(s_1 \rightarrow T_1)$ and $|T_2|$ -fan $A(s_2 \rightarrow T_2)$ with $s_1 \in V_b, s_2 \in V_w$ for $|T_1| + |T_2| = k$. Let $V(B(s_1 \rightarrow T_1, s_2 \rightarrow T_2))$ be the set of vertices incident with some paths in $B(s_1 \rightarrow T_1, s_2 \rightarrow T_2)$. A k^* -bifan $B(s_1 \rightarrow T_1, s_2 \rightarrow T_2)$ with different color sources of G is a k -fan with $V(B(s_1 \rightarrow T_1, s_2 \rightarrow T_2)) = V(G)$. A graph G is k^* -bifanable with different color sources if $\forall s_1 \in V_x, s_2 \in V_y$ and $\forall T_1 = \{t_1^1, t_1^2, \dots, t_1^{k_1} | t_1^1 \in V_y \text{ and } t_1^2, t_1^3, \dots, t_1^{k_1} \in V_x\}$ and $T_2 = \{t_2^1, t_2^2, \dots, t_2^{k_2} | t_2^1 \in V_x \text{ and } t_2^2, t_2^3, \dots, t_2^{k_2} \in V_y\}$ for $\{x, y\} = \{b, w\}$ and $k_1 + k_2 = k$, there exists a k^* -bifan. The authors proved that $Q_n - F_a$ is $(n - 1 - f_a)^*$ -bifanable with different color sources.

In this paper, we investigate the bifanability with two same color source vertices. A graph $G = (V_b \cup V_w, E)$ is k^* -bifanable with the same color sources, if $\forall s_1, s_2 \in V_x$ and $\forall T_1 = \{t_1^1, t_1^2, \dots, t_1^{k_1} | t_1^1 \in V_y \text{ and } t_1^2, t_1^3, \dots, t_1^{k_1} \in V_x\}$ and

*This is the footnote of this paper.

$T_2 = \{t_2^1, t_2^2, \dots, t_2^{k_2} | t_2^1 \in V_y \text{ and } t_2^2, t_2^3, \dots, t_2^{k_2} \in V_x\}$ for $\{x, y\} = \{b, w\}$ and $k_1 + k_2 = k$, there exists a k^* -bifan. In this paper, we will show that $Q_n - F_a$ is $(n - 1 - f_a)^*$ -bifanable with the same color sources if $f_a \leq n - 3$ where F_a is the set of f_a pairs of adjacently faulty vertices.

2 Preliminaries

An n -dimensional hypercube $Q_n = (V_b \cup V_w, E)$ is a bipartite graph whose vertices are labeled by distinct n -bit binary strings. Two vertices are linked by an edge if and only if their labels differ exactly in one bit. The hypercube Q_n can be constructed recursively as $Q_n = Q_{n-1} \times K_2$. We can partition Q_n as two subgraphs Q_n^0 and Q_n^1 by choosing any one bit of binary string.

We call the V_b black vertex set and V_w white vertex set. Let V_b^j and V_w^j be the black and white vertex set of Q_n^j , for $j = 0, 1$. And let $V^j = V_b^j \cup V_w^j$, for $j = 0, 1$. Thus, $V_b = V_b^0 \cup V_b^1, V_w = V_w^0 \cup V_w^1, V = V_b \cup V_w = V^0 \cup V^1$.

A bipartite graph $G = (V_b \cup V_w, E)$ has property 2H if it satisfies the following conditions. Let $a_1, a_2 \in V_w$ and $b_1, b_2 \in V_b$ be arbitrary four distinct vertices of G . There exist two spanning disjoint paths $P(a_1, b_1)$ and $P(a_2, b_2)$ of G . Let Q_n^0 and Q_n^1 be the $(n - 1)$ -dimensional subgraphs of Q_n . Let x be a vertex of Q_n^i , $\phi(x)$ be vertex of Q_n^j such that $(x, \phi(x)) \in E$ for $\{i, j\} = \{b, w\}$.

Let V^i be the vertex set of Q_n^i , for $i = 0, 1$. Let F_a be the set of f_a pairs of adjacently faulty vertices. Let $F_a^i = F_a \cap V^i$ and f_a^i be the number of pairs of adjacently vertices of F_a^i for $i = 0, 1$. Let T_j^i be the destination set of s_j in Q_n^i for $i = 0, 1, j = 1, 2$.

We need some previous results proved in other papers. The following lemma is introduced in [1].

Lemma 1 *The graph Q_n is f -adjacency $(n - 2 - f)$ edges Hamiltonian for $0 \leq f \leq (n - 2)$, f -adjacency $(n - 2 - f)$ edges Hamiltonian laceable for $0 \leq f \leq (n - 3)$, and f -adjacency $(n - 3 - f)$ edges hyper-Hamiltonian laceable for $0 \leq f \leq (n - 3)$.*

The following lemma is proved in [6].

Lemma 2 *The graph Q_n is f -adjacency $(n - 3 - f)$ edges property 2H, for $0 \leq f \leq n - 3, n \geq 3$.*

The following lemmas are proved in [4]

Lemma 3 *Let $s_1, t_1 \in V_w$ and $s_2, t_2 \in V_b$ be two pairs of fault-free vertices and F_a be the set with f_a pairs of adjacently faulty vertices of Q_n . There exist two spanning disjoint paths $P(s_1, t_1)$ and $P(s_2, t_2)$ of $Q_n - F_a$ for $f_a \leq n - 4, n \geq 4$.*

Lemma 4 *The graph Q_n is f_a -adjacency $(n - f_a)^*$ -fanable, for $0 \leq f_a \leq n - 3, n \geq 3$.*

Lemma 5 *The graph Q_n is f -adjacency k^* -bifanable with different color sources for $2 \leq k \leq n - f - 1, f \leq n - 3$.*

3 The adjacent vertices fault-tolerance of bifanability

In this section, we will prove the adjacent vertices fault-tolerance of bifanability with the same color sources of hypercube.

Theorem 1 *The graph $Q_n - F_a$ is $(n - |F_a| - 1)$ -bifanable for sources with the same color if $|F_a| \leq n - 3$ for $n \geq 3$.*

Proof: We will prove this theorem by induction on n . We can use s_1, s_2 to denote the source vertices for $s_1, s_2 \in V_b$. Let $T_1 = \{t_1^1, t_1^2, \dots, t_1^{k_1} | t_1^1 \in V_w \text{ and } t_1^2, t_1^3, \dots, t_1^{k_1} \in V_b\}$ and $T_2 = \{t_2^1, t_2^2, \dots, t_2^{k_2} | t_2^1 \in V_w \text{ and } t_2^2, t_2^3, \dots, t_2^{k_2} \in V_b\}$ be the sets of destinations of s_1 and s_2 , respectively. Without loss of generality, we can assume that $k_1 \geq k_2$.

Applying Lemma 2, we can obtain that this theorem is true for $f_a = n - 3$. When $n = 4$, we only need to construct the three spanning internally disjoint paths $P(s_1, t_1^1), P(s_1, t_1^2)$ and $P(s_2, t_2^1)$. Applying Lemma 5, we can construct these three spanning internally disjoint paths $P(s_1, t_1^1), P(s_1, t_1^2)$ and $P(t_2^1, s_2)$. Thus, this theorem is true for $n = 4$. By the symmetry of hypercube, we can assume that every pair of adjacently faulty vertices is in the same sub-cube. We will prove this theorem for $F_a \leq n - 4$ and $n \geq 5$ with the following cases.

Case 1 $s_1, s_2 \in Q_n^i$ for some $i = 0, 1$.

Without loss of generality, we can assume that $s_1, s_2 \in Q_n^0$.

Case 1.1 $|T^1| = 0$ and $|f_a^1| = 0$.

Let $T'_1 = T_1 - \{t_1^{k_1}\}$. By induction hypothesis, we can construct $(n - 2 - f_a)^*$ -bifan $B(s_1 \rightarrow T'_1, s_2 \rightarrow T_2)$ of $Q_n^0 - F_a^0$. Suppose the vertex $t_1^{k_1}$ is on the path

$P(s_1, t_1^i)$ for some $1 \leq i \leq k_1 - 1$. We can denote the path $P(s_1, t_1^{k_1})$ as $\langle s_1 \xrightarrow{P(s_1, t_1^{k_1})} t_1^{k_1} \rangle$. Applying Lemma 1, we can construct a Hamiltonian path $P(\phi(s_1), \phi(x))$ of Q_n^1 . We can construct the $B(s_1 \rightarrow T_1, s_2 \rightarrow T_2)$ from $B(s_1 \rightarrow T_1', s_2 \rightarrow T_2)$ by replacing $P(x, t_1^i)$ with $\langle s_1, \phi(s_1) \xrightarrow{P(\phi(s_1), \phi(x))} \phi(x), x \xrightarrow{P(x, t_1^i)} t_1^i \rangle$ and $P(s_1, t_1^{k_1})$ of $Q_n - F_a$. Suppose that $t_1^{k_1}$ is on the path $P(s_2, t_2^i)$ for some $1 \leq i \leq k_2$. We can denote the path $P(s_2, t_2^i)$ as $\langle s_2 \xrightarrow{P(s_2, t_2^i)} x_1^{k_1}, x_2 \xrightarrow{P(s_2, t_2^i)} t_2^i \rangle$. Applying Lemma 3, we can construct two spanning disjoint paths $P(\phi(s_1), \phi(t_1^{k_1}))$ and $P(\phi(x^1), \phi(x_2))$ of Q_n^1 . We can construct the $B(s_1 \rightarrow T_1, s_2 \rightarrow T_2)$ from $B(s_1 \rightarrow T_1', s_2 \rightarrow T_2)$ by replacing $P(s_2, t_2^i)$ with $\langle s_1, \phi(s_1) \xrightarrow{P(\phi(s_1), \phi(t_1^{k_1}))} \phi(t_1^{k_1}), t_1^{k_1} \rangle$ and $\langle s_2 \xrightarrow{P(s_2, x^1)} x^1, \phi(x^1) \xrightarrow{P(\phi(x^1), \phi(x_2))} \phi(x_2), x_2 \xrightarrow{P(x_2, t_2^i)} t_2^i \rangle$ of $Q_n - F_a$.

Case 1.2 $|T^1| = 0$ and $|F_a^1| \geq 1$.

By induction hypothesis, we can construct the $B(s_1 \rightarrow T_1, s_2 \rightarrow T_2)$ of $Q_n^0 - F_a^0$. Without loss of generality, we can assume that the path $P(s_1, t_1^1)$ can be denoted as $\langle s_1 \xrightarrow{P(s_1, x^1)} x^1, x^2 \xrightarrow{P(x^2, t_1^1)} t_1^1 \rangle$ for $\phi(x^1), \phi(x^2) \notin F_a^1$. Thus, we can construct $B(s_1 \rightarrow T_1, s_2 \rightarrow T_2)$ by replacing $P(s_1, t_1^1)$ with $\langle s_1 \xrightarrow{P(s_1, x^1)} x^1, \phi(x^1) \xrightarrow{P(\phi(x^1), \phi(x^2))} \phi(x^2), x^2 \xrightarrow{P(x^2, t_1^1)} t_1^1 \rangle$.

Case 1.3 $|T^1| \geq 1$ and $|F_a^1| \geq 1$.

Case 1.3.1 $|T^1| + |F_a^1| \leq n - 3$ and $|T^1| = 1$.

Without loss of generality, we can assume that $t_1^1 \in Q_n^1$. Let $x_1 \in Q_n^0$ be a white vertex such that $x^1, \phi(x^1) \notin (F_a^1 \cup T_1 \cup T_2)$. Applying Lemma 1, we can construct a Hamiltonian path $P(\phi(x^1), t_1^1)$ of $Q_n^1 - F_a^1$. By induction hypothesis, we can construct $B(s_1 \rightarrow \{x^1, t_1^2, \dots, t_1^{k_1}\}, s_2 \rightarrow T_2)$ of $Q_n^0 - F_a^0$. Thus, we can construct $B(s_1 \rightarrow T_1, s_2 \rightarrow T_2)$ of $Q_n - F_a$ from $B(s_1 \rightarrow \{x^1, t_1^2, \dots, t_1^{k_1}\}, s_2 \rightarrow T_2)$ by replacing $P(s_1, x^1)$ with

$$\langle s_1 \xrightarrow{P(s_1, x^1)} x_1, \phi(x^1) \xrightarrow{P(\phi(x^1), t_1^1)} t_1^1 \rangle.$$

Case 1.3.2 $|T^1| + |F_a^1| \leq n - 2$ and $|T^1| \geq 2$.

Without loss of generality, we can assume that $(t_1^1, t_1^2, \dots, t_1^{i_1}, t_2^1, t_2^2, \dots, t_2^{i_2}) \in Q_n^1$ and $i_1 \geq 2$. Let $(\phi(x^i), t_1^i)$ and $(\phi(y^j), t_2^j)$ be edge of Q_n^1 such that $x^i, y^j, \phi(x^i), \phi(y^j) \notin (F_a \cup T_1 \cup T_2 \cup \{s_1, s_2\})$ for $1 \leq i \leq k_1, 1 \leq j \leq k_2$. Applying Lemma 2, we can construct two spanning disjoint paths $P(\phi(x^1), t_1^1)$ and $P(\phi(y^1), t_2^1)$. By induction hypothesis, we can construct $B(s_1 \rightarrow \{x^1, \dots, x^{k_1}\}, s_2 \rightarrow \{y^1, \dots, y^{k_2}\})$ of $Q_n^0 - F_a^0$. Thus, the paths $\langle s_1 \xrightarrow{P(s_1, x^1)} x^1, \phi(x^1) \rangle$, $\langle s_1 \xrightarrow{P(s_1, x^i)} x^i, \phi(x^i) \xrightarrow{P(\phi(x^i), t_1^i)} t_1^i \rangle$, $\langle s_2 \xrightarrow{P(s_2, y^j)} y^j, \phi(y^j), t^j \rangle$, $\langle s_2 \xrightarrow{P(s_2, y^1)} y^1, \phi(y^1), t_2^1 \rangle$ form the $(n - 1 - f_a)^*$ -bifan of $Q_n - F_a$ for $2 \leq i \leq k_1, 2 \leq j \leq k_2$.

Case 1.3.3 $|T^1| \geq 2$ and $|T^1| + |F_a^1| = 1$.

Let $(\phi(x^i), t_1^i)$ and $(\phi(y^j), t_2^j)$ be edges of Q_n^1 such that $x^i, y^j, \phi(x^i), \phi(y^j) \notin (F_a \cup T_1 \cup T_2 \cup \{s_1, s_2\})$ for $3 \leq i \leq k_1, 2 \leq j \leq k_2$. Applying Lemma 1, we can construct a hyper-Hamiltonian laceable $\langle t_1^1 \xrightarrow{P(t_1^1, x^1)} x^1, y^1 \xrightarrow{P(y^1, t_2^1)} t_2^1 \rangle$ and $(\phi(x^2), t_1^2)$ be a edge of Q_n . By induction hypothesis, we can construct $B(s_1 \rightarrow \{x^1, \phi(t_1^2), x^3, \dots, x^{k_1}\}, s_2 \rightarrow \{y^1, y^2, \dots, y^{k_2}\})$. Thus, the paths $\langle s_1 \xrightarrow{P(s_1, x^1)} x^1, \phi(x^1) \xrightarrow{P(\phi(x^1), t_1^1)} t_1^1 \rangle$, $\langle s_1 \xrightarrow{P(s_1, \phi(t_1^2))} \phi(t_1^2), t_1^2 \rangle$, $\langle s_1 \xrightarrow{P(s_1, x^i)} x^i, \phi(x^i), t_1^i \rangle$, $\langle s_2 \xrightarrow{P(s_2, y^1)} y^1, \phi(y^1) \xrightarrow{P(\phi(y^1), t_2^1)} t_2^1 \rangle$, $\langle s_2 \xrightarrow{P(s_2, y^j)} y^j, \phi(y^j), t_2^{k_2} \rangle$ form the $(n - 1 - f_a)^*$ -bifan of $Q_n - F_a$ for $3 \leq i \leq k_1, 2 \leq j \leq k_2$.

Case 1.4 $|T^1| \geq 1$ and $|F_a^1| = 0$.

Case 1.4.1 $t_1^1, t_2^1 \notin T^1$ and $|T^1| \geq 1$.

Without loss of generality, we can assume that $T^1 = \{t_1^{i_1}, \dots, t_1^{k_1}, t_2^{i_2}, \dots, t_2^{k_2}\}$ for $2 \leq i_1 \leq k_1, 2 \leq i_2 \leq k_2$. Let $(\phi(x^i), t_1^i), (\phi(y^j), t_2^j) \in Q_n^1$ for $x^i, y^j \notin (\{s_1, s_2\} \cup F_a)$ for $i_1 \leq i \leq k_1 - 1, i_2 \leq j \leq k_2$. Applying Lemma 1, we can construct a Hamiltonian path $P(\phi(s_1), t_1^{k_1})$

of $Q_n^1 - \{x^i, t_1^i, y^j, t_2^j | i_1 \leq i \leq k_1 - 1, i_2 \leq j \leq k_2\}$. By induction hypothesis, we can construct $B(s_1 \rightarrow \{t_1^1, \dots, t_1^{i_1-1}, x^{i_1}, \dots, x_1^{k_1-1}\}, s_2 \rightarrow \{t_2^1, \dots, t_2^{i_2-1}, y^{i_2}, \dots, y^{k_2}\})$.

Thus, the paths $P(s_1, t_1^{j_1}), \langle s_1 \xrightarrow{P(s_1, x^{j_2})} x^{j_2}, \phi(x^{j_2}), t_1^{j_2} \rangle, \langle s_1, \phi(s_1) \xrightarrow{P(\phi(s_1), t_1^{k_1})} t_1^{k_1} \rangle, P(s_2, t_2^{j_3})$ and $\langle s_2 \xrightarrow{P(s_2, y^{j_4})} y^{j_4}, \phi(y^{j_4}), t_2^{j_4} \rangle$ form the $(n - 1 - f_a)^*$ -bifan of $Q_n - F_a$ for $1 \leq j_1 \leq i_1 - 1, i_1 \leq j_2 \leq k_1 - 1, 1 \leq j_3 \leq i_2 - 1, i_2 \leq j_4 \leq k_2$.

Case 1.4.2 $T^1 = \{t_1^1\}$ or $T^1 = \{t_2^1\}$.

Without loss of generality, we can assume that $T^1 = \{t_1^1\}$. Let $(x^1, t_1^{k_1}) \in Q_n^0$ for $x^1 \notin (\{t_2^1\} \cup F_a)$. By induction hypothesis, we can construct $B(s_1 \rightarrow \{x^1, t_2^1, \dots, t_1^{k_1-1}\}, s_2 \rightarrow \{t_2^1, \dots, t_2^{k_2}\})$. Without loss of generality, we can assume that $t_1^{k_1}$ is on the path $P(s_2, t_2^1)$.

We can denote this path as $\langle s_2 \xrightarrow{P(s_2, a)} a, t_1^{k_1}, b \xrightarrow{P(b, t_2^1)} t_2^1 \rangle$.

Suppose $\phi(s_1) \neq t_1^1$. Applying Lemma 3, we can construct two paths $P(\phi(s_1), t_1^1)$ and $P(\phi(a), \phi(b))$ of Q_n^1 .

Thus, the paths $\langle s_1, \phi(s_1) \xrightarrow{P(\phi(s_1), t_1^1)} t_1^1 \rangle, P(s_1, t_1^i), \langle s_1 \xrightarrow{P(s_1, x^1)} t_1^{k_1} \rangle, \langle s_2 \xrightarrow{P(s_2, a)} a, \phi(a) \xrightarrow{P(\phi(a), \phi(b))} \phi(b), b \xrightarrow{P(b, t_2^1)} t_2^1 \rangle, P(s_2, t_2^j)$ form the $(n - 1 - f_a)^*$ -bifan of $Q_n - F_a$ for $2 \leq i \leq k_1 - 1, 2 \leq j \leq k_2$.

Suppose $\phi(s_1) = t_1^1$. Applying Lemma 1, we can construct a Hamiltonian path $P(\phi(a), \phi(b)) \in Q_n^1 - \{t_1^1\}$. Thus, the paths $\langle s_1, t_1^1 \rangle, P(s_1, t_1^i), \langle s_1 \xrightarrow{P(s_1, x^1)} t_1^{k_1} \rangle, \langle s_2 \xrightarrow{P(s_2, a)} a, \phi(a) \xrightarrow{P(\phi(a), \phi(b))} \phi(b), b \xrightarrow{P(b, t_2^1)} t_2^1 \rangle, P(s_2, t_2^j)$ form the $(n - 1 - f_a)^*$ -bifan of $Q_n - F_a$ for $2 \leq i \leq k_1 - 1, 2 \leq j \leq k_2$, as illustrated in Figure 1. (a).

Case 1.4.3 exactly one vertex of t_1^1, t_1^1 in T^1 and $|T^1| \geq 2$.

Without loss of generality, we can assume that $T^1 = \{t_1^1, \dots, t_1^{i_1}, t_2^1, \dots, t_2^{i_2}\}$ for $1 \leq i_1 \leq k_1, 2 \leq i_2 \leq k_2$. Let $(\phi(x^i), t_1^i), (\phi(y^j), t_2^j) \in Q_n^1$ for $x^i, y^j \notin (\{s_1, s_2\} \cup F_a)$ for $1 \leq i \leq i_1 - 1, i_2 \leq j \leq k_2$. Let $x^1 \in V_w^0$ for $x^1 \notin \{t_2^1\} \cup F_a$ and

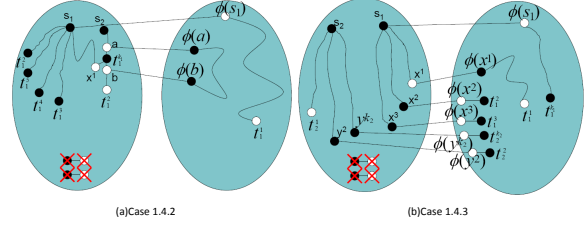


Figure 1: The case 1.4.2 and case 1.4.3 of Theorem 1.

$\phi(x^1) \notin T^1$. By induction hypothesis, we can construct $B(s_1 \rightarrow \{x^1, \dots, x^{i_1-1}, t_1^{i_1+1}, \dots, t_1^{k_1}\}, s_2 \rightarrow \{t_2^1, \dots, t_2^{i_2-1}, y^{i_2}, \dots, y^{k_2}\})$.

Suppose $\phi(s_1) \neq t_1^1$. Applying Lemma 2, we can construct two paths $P(\phi(s_1), t_1^1), P(\phi(x^1), t_1^1)$

of $Q_n^1 - \{t_1^{j_1}, \phi(x^{j_1}), t_2^{j_2}, \phi(y^{j_2}) | 2 \leq j_1 \leq i_1 - 1, i_2 \leq j_2 \leq k_2\}$. Thus, the paths $\langle s_1 \xrightarrow{P(s_1, x^1)} x^1, \phi(x^1) \xrightarrow{P(\phi(x^1), t_1^1)} t_1^1 \rangle, \langle s_1 \xrightarrow{P(s_1, x^{j_1})} x^{j_1}, \phi(x^{j_1}), t_1^{j_1} \rangle, \langle s_1, \phi(s_1) \xrightarrow{P(\phi(s_1), t_1^1)} t_1^1 \rangle, P(s_1, t_1^{j_2}),$

$P(s_2, t_2^{j_3}), \langle s_2 \xrightarrow{P(s_2, y^{j_4})} y^{j_4}, \phi(y^{j_4}), t_2^{j_4} \rangle$ form the $(n - 1 - f_a)^*$ -bifan of $Q_n - F_a$ for $2 \leq j_1 \leq i_1 - 1, i_1 + 1 \leq j_2 \leq k_1, 2 \leq j_3 \leq i_2 - 1, i_2 \leq j_4 \leq k_2$.

Suppose $\phi(s_1) = t_1^1$. Applying Lemma 1, we can construct a Hamiltonian path $P(\phi(x^1), t_1^1)$ of $Q_n^1 - \{t_1^{j_1}, \phi(x^{j_1}), t_2^{j_2}, \phi(y^{j_2}) | 2 \leq j_1 \leq i_1 - 1, i_2 \leq j_2 \leq k_2\} - \{t_1^1\}$. Thus, the paths $\langle s_1, t_1^1 \rangle, \langle s_1 \xrightarrow{P(s_1, x^{j_1})} x^{j_1}, \phi(x^{j_1}), t_1^{j_1} \rangle, \langle s_1 \xrightarrow{P(s_1, x^1)} x^1, \phi(x^1) \xrightarrow{P(\phi(x^1), t_1^1)} t_1^1 \rangle, P(s_1, t_1^{j_2}), P(s_2, t_2^{j_3}), \langle s_2 \xrightarrow{P(s_2, y^{j_4})} y^{j_4}, \phi(y^{j_4}), t_2^{j_4} \rangle$ form the $(n - 1 - f_a)^*$ -bifan of $Q_n - F_a$ for $2 \leq j_1 \leq i_1 - 1, i_1 + 1 \leq j_2 \leq k_1, 2 \leq j_3 \leq i_2 - 1, i_2 \leq j_4 \leq k_2$, as illustrated in Figure 1. (b).

Case 1.4.4 $|T^1| = \{t_1^1, t_2^1\}$.

When $n = 5$, we can distribute s_2, t_1^1 into different sub-cubes which will be discussed in other cases. Thus, we can assume that $n \geq 6$. Let $(\phi(y^1), t_2^1) \in Q_n^1$ for $y^1 \in (Q_n^0 - F_a)$. Let $(x^1, t_1^{k_1}) \in Q_n^0$ be an edge such that $x^1 \notin (F_a \cup \{y^1\})$. By induction hypothesis, we can construct $B(s_1 \rightarrow \{x^1, t_1^2, \dots, t_1^{k_1-1}\}, s_2 \rightarrow$

$\{y^1, t_2^2, \dots, t_2^{k_2}\}$). Without loss of generality, we can assume that $t_1^{k_1}$ is on the path $P(s_1, t_1^2)$. We can denote this path as $\langle s_1 \xrightarrow{P(s_1, a)} a, t_1^{k_1}, b \xrightarrow{P(b, t_1^2)} t_1^2 \rangle$. Suppose $\phi(s_1) \neq t_1^1$. Applying Lemma 3, we can construct two spanning disjoint paths $P(\phi(s_1), t_1^1)$, $P(\phi(a), \phi(b))$ of $Q_n^1 - \{t_2^1, \phi(y^1)\}$. Thus, the paths $\langle s_1, \phi(s_1) \xrightarrow{P(\phi(s_1), t_1^1)} t_1^1 \rangle$, $\langle s_1 \xrightarrow{P(s_1, a)} a, \phi(a) \xrightarrow{P(\phi(a), \phi(b))} \phi(b), b \xrightarrow{P(b, t_1^2)} t_1^2 \rangle$, $P(s_1, t_1^{j_1})$, $\langle s_1 \xrightarrow{P(s_1, x^1)} x^1, t_1^{k_1} \rangle$, $\langle s_2 \xrightarrow{P(s_2, y^1)} y^1, \phi(y^1), t_2^1 \rangle$, $P(s_2, t_2^{j_2})$ form the $(n-1-f_a)^*$ -bifan of $Q_n - F_a$ for $3 \leq j_1 \leq k_1-1, 2 \leq j_2 \leq k_2$. Suppose $\phi(s_1) = t_1^1$. Applying Lemma 1, we can construct a Hamiltonian path $P(\phi(a), \phi(b)) \in (Q_n^1 - \{t_1^1, t_2^1, \phi(y^1)\})$. Thus, the paths $\langle s_1, t_1^1 \rangle$, $\langle s_1 \xrightarrow{P(s_1, a)} a, \phi(a) \xrightarrow{P(\phi(a), \phi(b))} \phi(b), b \xrightarrow{P(b, t_1^2)} t_1^2 \rangle$, $P(s_1, t_1^{j_1})$, $\langle s_1 \xrightarrow{P(s_1, x^1)} x^1, t_1^{k_1} \rangle$, $\langle s_2 \xrightarrow{P(s_2, y^1)} y^1, \phi(y^1), t_2^1 \rangle$, $P(s_2, t_2^{j_2})$ form the $(n-1-f_a)^*$ -bifan of $Q_n - F_a$ for $3 \leq j_1 \leq k_1-1, 2 \leq j_2 \leq k_2$, as illustrated in Figure 2. (a).

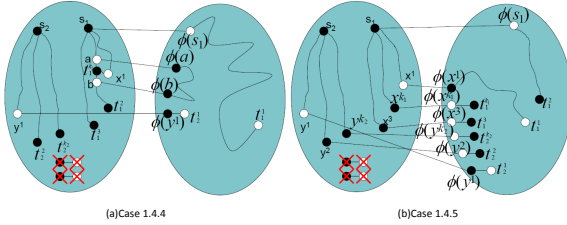


Figure 2: The case 1.4.4 and case 1.4.5 of Theorem 1.

Case 1.4.5 $t_1^1, t_2^1 \in T^1$ and $3 \leq |T^1| \leq n-2$. Without loss of generality, we can assume that $T^1 = \{t_1^1, \dots, t_1^{i_1}, t_2^1, \dots, t_2^{i_2}\}$ for $1 \leq i_1 \leq k_1, 1 \leq i_2 \leq k_2$. Let $(\phi(x^i), t_1^i), (\phi(y^j), t_2^j) \in Q_n^1$ for $x^i, y^j \notin (\{s_1, s_2\} \cup F_a \cup T^0)$ and $\phi(x^i), \phi(y^j) \notin (T^1 \cup \{\phi(s_1)\})$, $3 \leq i \leq i_1, 1 \leq j \leq i_2$. Let x^1 be a white vertex of $Q_n^0 - (T^0 \cup F_a \cup \{s_1, s_2\})$ such that $\phi(x^1) \notin (T^1 \cup \{\phi(y^1)\})$. By induction hypothesis, we can construct $B(s_1 \rightarrow \{x^1, \dots, x^{i_1}, t_1^{i_1+1}, \dots, t_1^{k_1}\}, s_2 \rightarrow$

$\{y^1, \dots, y^{i_2}, t_2^{i_2+1}, \dots, t_2^{k_2}\})$. Suppose $\phi(s_1) \neq t_1^1$. Applying Lemma 2, we can construct two spanning disjoint paths $P(\phi(s_1), t_1^2)$ and $P(\phi(x^1), \phi(t_1^1))$ of $Q_n^1 - \{t_1^{j_1}, \phi(x^{j_1}), t_2^{j_2}, \phi(y^{j_2}) \mid 3 \leq j_1 \leq i_1, 1 \leq j_2 \leq i_2\}$. Thus, the paths $\langle s_1 \xrightarrow{P(s_1, x^1)} x^1, \phi(x^1) \xrightarrow{P(\phi(x^1), t_1^1)} t_1^1 \rangle$, $\langle s_1, \phi(s_1) \xrightarrow{P(\phi(s_1), \phi(t_1^2))} t_1^2 \rangle$, $\langle s_1 \xrightarrow{P(s_1, x^{j_1})} x^{j_1}, \phi(x^{j_1}), t_1^{j_1} \rangle$, $P(s_1, t_1^{j_2})$, $\langle s_2 \xrightarrow{P(s_2, y^{j_3})} y^{j_3}, \phi(y^{j_3}), t_2^{j_3} \rangle$, $P(s_2, t_2^{j_4})$ form the $(n-1-f_a)^*$ -bifan of $Q_n - F_a$ for $3 \leq j_1 \leq i_1, i_1+1 \leq j_2 \leq k_1, 1 \leq j_3 \leq i_2, i_2+1 \leq j_4 \leq k_2$. Suppose $\phi(s_1) = t_1^1$. Applying Lemma 1, we can construct a Hamiltonian path $P(\phi(x^1), \phi(t_1^2))$ in $Q_n^1 - \{t_1^{j_1}, \phi(x^{j_1}), t_2^{j_2}, \phi(y^{j_2}) \mid 3 \leq j_1 \leq i_1, 1 \leq j_2 \leq i_2\} - \{t_1^1\}$. Thus, the paths $\langle s_1, t_1^1 \rangle$, $\langle s_1 \xrightarrow{P(s_1, x^1)} x^1, \phi(x^1) \xrightarrow{P(\phi(x^1), t_1^2)} t_1^2 \rangle$, $\langle s_1 \xrightarrow{P(s_1, x^{j_1})} x^{j_1}, \phi(x^{j_1}), t_1^{j_1} \rangle$, $P(s_1, t_1^{j_2})$, $\langle s_2 \xrightarrow{P(s_2, y^{j_3})} y^{j_3}, \phi(y^{j_3}), t_2^{j_3} \rangle$, $P(s_2, t_2^{j_4})$ form the $(n-1-f_a)^*$ -bifan of $Q_n - F_a$ for $3 \leq j_1 \leq i_1, i_1+1 \leq j_2 \leq k_1, 1 \leq j_3 \leq i_2, i_2+1 \leq j_4 \leq k_2$, as illustrated in Figure 2. (b).

Case 1.4.6 $|T^1| = n-1$.

Let $(\phi(x^i), t_1^i), (\phi(y^j), t_2^j) \in Q_n^1$ such that $x^i, y^j \notin \{s_1, s_2\}$ and $\phi(x^i), \phi(y^j) \notin \{t_1^1, t_2^1\}$ for $3 \leq i \leq k_1, 2 \leq j \leq k_2$. Applying Lemma 1, we can construct a Hamiltonian path $P(t_1^2, t_2^1)$ of $Q_n^1 - \{\phi(x^i), t_1^i, \phi(y^j), t_2^j \mid 3 \leq i \leq k_1, 2 \leq j \leq k_2\}$. Without loss of generality, we can denote $P(t_1^2, t_2^1)$ as $\langle t_1^2 \xrightarrow{P(t_1^2, \phi(x^2))} \phi(x^2), t_1^1 \xrightarrow{P(t_1^1, \phi(s_1))} \phi(s_1), \phi(y^1) \xrightarrow{P(\phi(y^1), t_2^1)} t_2^1 \rangle$. By induction hypothesis, we can construct $B(s_1 \rightarrow \{x^2, \dots, x^{k_1}\}, s_2 \rightarrow \{y^1, \dots, y^{k_2}\})$. Thus, the paths $\langle s_1, \phi(s_1) \xrightarrow{P(\phi(s_1), t_1^1)} t_1^1 \rangle$, $\langle s_1 \xrightarrow{P(s_1, x^2)} x^2, \phi(x^2) \xrightarrow{P(\phi(x^2), t_1^1)} t_1^1 \rangle$, $\langle s_1 \xrightarrow{P(s_1, x^{j_1})} x^{j_1}, \phi(x^{j_1}), t_1^{j_1} \rangle$, $\langle s_2 \xrightarrow{P(s_2, y^1)} y^1, \phi(y^1) \xrightarrow{P(\phi(y^1), t_2^1)} t_2^1 \rangle$, $\langle s_2 \xrightarrow{P(s_2, y^{j_2})} y^{j_2}, \phi(y^{j_2}), t_2^{j_2} \rangle$ form the $(n-1-f_a)^*$ -bifan of $Q_n - F_a$ for $3 \leq j_1 \leq k_1, 2 \leq j_2 \leq k_2$, as illustrated in Figure 3. (a).

Case 2 $s_1 \in Q_n^i$ and $s_2 \in Q_n^j$ for $\{i, j\} = \{0, 1\}$.

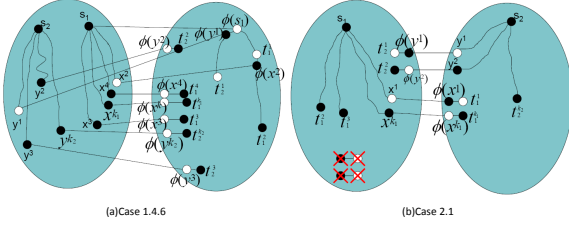


Figure 3: The case 1.4.6 and case 2.1 of Theorem 1.

Without loss generality, we can assume that $s_1 \in Q_n^0, s_2 \in Q_n^1$. We can also assume that $|T_1^1| + |F_a^1| \geq |T_2^0| + |F_a^0|$.

Case 2.1 $|T_2^0| + |F_a^0| \leq |T_1^1| + |F_a^1| \leq n - 4$.

Without loss generality, we can assume that $t_1^1, \dots, t_{i_1}^1 \in Q_n^1$ and $t_2^0, \dots, t_{i_2}^0 \in Q_n^0$. Let $(\phi(x^i), t_1^1) \in Q_n^1$ for $x^i \notin (F_a^0 \cup T^0)$ and $\phi(x^i) \notin (F_a^1 \cup T^1)$ for $1 \leq i \leq i_1$. Let $(\phi(y^j), t_2^0) \in Q_n^0$ for $y^j \notin (F_a^1 \cup T^1)$ and $\phi(y^j) \notin (F_a^0 \cup T^0)$ for $1 \leq j \leq i_2$. Applying Lemma 4, we can construct a k_1^* -fan $A(s_1 \rightarrow \{x^1, \dots, x^{i_1}, t_1^{i_1+1}, \dots, t_1^{k_1}\})$ of $Q_n^0 - F_a^0 - \{\phi(y^j), t_2^j \mid 1 \leq j \leq i_2\}$ and a k_2^* -fan $A(s_2 \rightarrow \{y^1, \dots, y^{i_2}, t_2^{i_2+1}, \dots, t_2^{k_2}\})$ of $Q_n^1 - F_a^1 - \{\phi(x^i), t_1^i \mid 1 \leq i \leq i_1\}$. Thus, the following paths $\langle s_1 \xrightarrow{P(s_1, x^{j_1})} x^{j_1}, \phi(x^{j_1}), t_1^{j_1} \rangle, P(s_1, t_1^{j_2})$ and $\langle s_2 \xrightarrow{P(s_2, \phi(y^{j_3}))} \phi(y^{j_3}), y^{j_3}, t_2^{j_3} \rangle, P(s_2, t_2^{j_4})$ form the $(n - 1 - f_a)^*$ -bifan of $Q_n - F_a$ for $1 \leq j_1 \leq i_1, i_1 + 1 \leq j_2 \leq k_1, 1 \leq j_3 \leq i_2, i_2 + 1 \leq j_4 \leq k_2$, as illustrated in Figure 3. (b).

Case 2.2 $|T_2^0| + |F_a^0| \leq |T_1^1| + |F_a^1| = n - 3$.

Since $|T_1^1| + |F_a^1| = n - 3, |T_2| \leq 2$. Suppose that $|T_2^0| + |F_a^0| = |T_1^1| + |F_a^1| = n - 3$. We can obtain that $n = 5$ and $f_a \leq 1$. We can distribute s_2 and t_1^2 into different sub-cubes. In the following, we will discuss that $|T_2^0| + |F_a^0| \leq n - 4$.

Case 2.2.1 $|T_2| = 2$

We can obtain that $(T_1 \cup F_a) \subset Q_n^1$.

Case 2.2.1.1 $t_1^1, t_2^2 \in Q_n^1$.

Let $(\phi(x^i), t_1^1) \in Q_n^1$ such that $x^i \notin \{s_1\}$ and $\phi(x^i) \notin (F_a \cup T_2 \cup \{s_2, t_1^{k_1}\})$ for $1 \leq i \leq k_1 - 1$. Applying Lemma 4, we can construct a 3^* -fan

$P(s_2, t_2^1), P(s_2, t_2^2), \langle s_2, \phi(x^{k_1}) \xrightarrow{P(\phi(x^{k_1}), t_1^{k_1})} t_1^{k_1} \rangle$ of $Q_n^1 - F_a - \{t_1^i, \phi(x^i) \mid 1 \leq i \leq k_1 - 1\}$

$i \leq k_1 - 1\}$. Applying Lemma 4, we also can construct the k_1^* -fan $A(s_1 \rightarrow \{x^1, \dots, x^{k_1}\})$ of Q_n^0 . Thus, the paths $\langle s_1 \xrightarrow{P(s_1, x^i)} x^i, \phi(x^i), t_1^i \rangle, \langle s_1 \xrightarrow{P(s_1, x^{k_1})} x^{k_1}, \phi(x^{k_1}) \xrightarrow{P(\phi(x^{k_1}), t_1^{k_1})} t_1^{k_1} \rangle, P(s_2, t_2^1)$ and $P(s_2, t_2^2)$ form the $(n - 1 - f_a)^*$ -bifan of $Q_n - F_a$ for $1 \leq i \leq k_1 - 1$, as illustrated in Figure 4. (a).

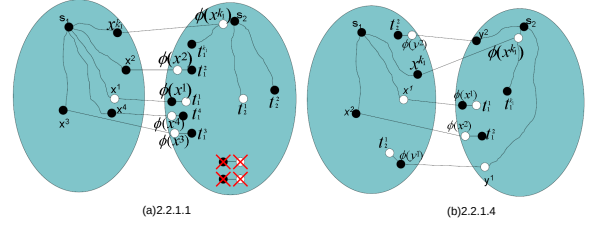


Figure 4: The case 2.2.1.1 and 2.2.1.4 of Theorem 1.

Case 2.2.1.2 $t_1^1 \in Q_n^0$ and $t_2^2 \in Q_n^1$.

Let $(\phi(x^i), t_1^1) \in Q_n^1$ such that $x^i \notin \{s_1, t_2^2\}$ and $\phi(x^i) \notin (F_a \cup \{s_2, t_2^2, t_1^{k_1}\})$ for $1 \leq i \leq k_1 - 1$. Let $(\phi(y^1), t_2^1) \in Q_n^0$ such that $y^1 \notin (F_a \cup T_1 \cup \{\phi(x^1), \dots, \phi(x^{k_1-1})\})$ and $\phi(y^1) \notin \{x^1, \dots, x^{k_1-1}\}$. Applying Lemma 4, we can construct a 3^* -fan $P(s_2, t_2^2), P(s_2, y^1),$

$\langle s_2, \phi(x^{k_1}) \xrightarrow{P(\phi(x^{k_1}), t_1^{k_1})} t_1^{k_1} \rangle$ of $Q_n^1 - F_a - \{t_1^i, \phi(x^i) \mid 1 \leq i \leq k_1 - 1\}$. Applying Lemma 4, we can construct a k_1^* -fan $A(s_1 \rightarrow \{x^1, x^2, \dots, x^{k_1}\})$ of $Q_n^0 - \{t_2^1, \phi(y^1)\}$. Thus, the

paths $\langle s_1 \xrightarrow{P(s_1, x^i)} x^i, \phi(x^i), t_1^i \rangle, \langle s_1 \xrightarrow{P(s_1, x^{k_1})} x^{k_1}, \phi(x^{k_1}) \xrightarrow{P(\phi(x^{k_1}), t_1^{k_1})} t_1^{k_1} \rangle, \langle s_2 \xrightarrow{P(s_2, y^1)} y^1, \phi(y^1), t_2^1 \rangle, P(s_2, t_2^2)$ form the $(n - 1 - f_a)^*$ -bifan of $Q_n - F_a$ for $1 \leq i \leq k_1 - 1$.

Case 2.2.1.3 $t_2^2 \in Q_n^0$ and $t_1^1 \in Q_n^1$.

Let $(\phi(x^i), t_1^1) \in Q_n^1$ such that $x^i \notin \{s_1, t_2^2\}$ and $\phi(x^i) \notin (F_a \cup \{s_2, t_2^1, t_1^{k_1}\})$ for $1 \leq i \leq k_1 - 1$.

Suppose that $d(s_2, t_2^2) > 2$. Let $(\phi(y^2), t_2^2) \in Q_n^0$ such that $y^2 \notin (F_a \cup \{\phi(x^1), \dots, \phi(x^{k_1-1})\} \cup T_1)$ and $\phi(y^2) \notin \{x^1, \dots, x^{k_1-1}\}$. Applying Lemma 4, we can construct a 3^* -fan $P(s_2, t_2^1), P(s_2, y^2),$

$\langle s_2, \phi(x^{k_1}) \xrightarrow{P(\phi(x^{k_1}), t_1^{k_1})} t_1^{k_1} \rangle$ of $Q_n^1 - F_a - \{t_1^i, \phi(x^i) | 1 \leq i \leq k_1 - 1\}$. Applying Lemma 4, we can construct a k_1^* -fan $A(s_1 \rightarrow \{x^1, x^2, \dots, x^{k_1}\})$ of $Q_n^0 - \{t_2^2, \phi(y^2)\}$. Thus, the paths $\langle s_1 \xrightarrow{P(s_1, x^i)} x^i, \phi(x^i), t_1^i \rangle$, $\langle s_1 \xrightarrow{P(s_1, x^{k_1})} x^{k_1}, \phi(x^{k_1}) \xrightarrow{P(\phi(x^{k_1}), t_1^{k_1})} t_1^{k_1} \rangle$, $\langle s_2, \phi(s_2), t_2^2 \rangle$ form the $(n-1-f_a)^*$ -bifan of $Q_n - F_a$ for $1 \leq i \leq k_1 - 1$.

Suppose $d(s_2, t_2^2) = 2$. Let x^{k_1} be a black vertex of Q_n^0 such that $x^{k_1} \notin \{s_1, t_2^2, x^1, \dots, x^{k_1-1}\}$ and $\phi(x^{k_1}) \notin (F_a \cup T_1 \cup \{t_2^1, \phi(x^1), \dots, \phi(x^{k_1-1})\})$. Applying Lemma 2, we can construct two spanning disjoint paths $P(\phi(x^{k_1}), t_1^{k_1}), P(s_2, t_2^1)$ of $Q_n^1 - (F_a \cup T_1 \cup \{\phi(x^1), \dots, \phi(x^{k_1-1})\})$. Applying Lemma 4, we can construct a k_1^* -fan $A(s_1 \rightarrow \{x^1, \dots, x^{k_1}\})$ of $Q_n^0 - \{\phi(s_2), t_2^2\}$. Thus, the paths $\langle s_1 \xrightarrow{P(s_1, x^i)} x^i, \phi(x^i), t_1^i \rangle$, $\langle s_1 \xrightarrow{P(s_1, x^{k_1})} x^{k_1}, \phi(x^{k_1}) \xrightarrow{P(\phi(x^{k_1}), t_1^{k_1})} t_1^{k_1} \rangle$, $\langle s_2, \phi(s_2), t_2^2 \rangle$ form the $(n-1-f_a)^*$ -bifan of $Q_n - F_a$ for $1 \leq i \leq k_1 - 1$.

Case 2.2.1.4 $t_2^1, t_2^2 \in Q_n^0$.

Suppose $d(s_2, t_2^2) \geq 3$. Let $(\phi(x^i), t_1^i) \in Q_n^1$ such that $x^i \notin \{s_1, t_2^1, t_2^2\}$ and $\phi(x^i) \notin (F_a \cup T_1 \cup \{s_2\})$ for $1 \leq i \leq k_1 - 1$. Let $(\phi(y^1), t_2^1), (\phi(y^2), t_2^2) \in Q_n^0$ such that $y^1, y^2 \notin (F_a \cup T_1 \cup \{s_2, \phi(x^1), \dots, \phi(x^{k_1-1})\})$ and $\phi(y^1), \phi(y^2) \notin \{s_1, x^1, \dots, x^{k_1-1}\}$. Applying Lemma 4, we can construct a 3^* -fan $P(s_2, y^1), P(s_2, y^2)$, $\langle s_2, \phi(x^{k_1}) \xrightarrow{P(\phi(x^{k_1}), t_1^{k_1})} t_1^{k_1} \rangle$ of $Q_n^1 - F_a - \{t_1^i, \phi(x^i) | 1 \leq i \leq k_1 - 1\}$. Applying Lemma 4, we can construct a k_1^* -fan $A(s_1 \rightarrow \{x^1, \dots, x^{k_1}\})$ of $Q_n^0 - \{\phi(y_1), t_2^1, \phi(y_2), t_2^2\}$. Thus, the paths $\langle s_1 \xrightarrow{P(s_1, x^i)} x^i, \phi(x^i), t_1^i \rangle$, $\langle s_1 \xrightarrow{P(s_1, x^{k_1})} x^{k_1}, \phi(x^{k_1}) \xrightarrow{P(\phi(x^{k_1}), t_1^{k_1})} t_1^{k_1} \rangle$, $\langle s_2 \xrightarrow{P(s_2, y^1)} y^1, \phi(y^1), t_2^1 \rangle$, $\langle s_2 \xrightarrow{P(s_2, y^2)} y^2, \phi(y^2), t_2^2 \rangle$ form the $(n-1-f_a)^*$ -bifan of $Q_n - F_a$ for $1 \leq i \leq k_1 - 1$, as illustrated in Figure 4. (b).

Suppose $d(s_2, t_2^2) = 2$ and $d(s_2, t_2^1) \geq 2$. Let $\phi(x^i, t_1^i) \in Q_n^1$ such that $x^i \notin (T_2 \cup \{s_1\})$ and $\phi(x^i) \notin (F_a \cup$

$T_1 \cup \{s_2\})$ for $1 \leq i \leq k_1$. Let $(\phi(y^1), t_2^1) \in Q_n^0$ such that $y^1 \notin (T_1 \cup F_a \cup \{\phi(x^1), \dots, \phi(x^{k_1-1})\})$ and $\phi(y^1) \notin \{s_1, t_2^2, x^1, \dots, x^{k_1-1}\}$. Applying Lemma 2, we can construct two spanning disjoint paths $P(x^{k_1}, t_1^{k_1})$ and $P(s_2, y^1)$ of $Q_n^1 - (F_a \cup T_1 \cup \{\phi(x^1), \dots, \phi(x^{k_1-1})\})$. Applying Lemma 4, we can construct a k_1^* -fan $A(s_1 \rightarrow \{x^1, \dots, x^{k_1}\})$ of $Q_n^0 - \{t_2^1, t_2^2, \phi(y^1), \phi(s_2)\}$. Thus, the paths $\langle s_1 \xrightarrow{P(s_1, x^i)} x^i, \phi(x^i), t_1^i \rangle$, $\langle s_1 \xrightarrow{P(s_1, x^{k_1})} x^{k_1}, \phi(x^{k_1}) \xrightarrow{P(\phi(x^{k_1}), t_1^{k_1})} t_1^{k_1} \rangle$, $\langle s_2 \xrightarrow{P(s_2, y^1)} y^1, \phi(y^1), t_2^1 \rangle$, $\langle s_2, \phi(s_2), t_2^2 \rangle$ form the $(n-1-f_a)^*$ -bifan of $Q_n - F_a$ for $1 \leq i \leq k_1 - 1$, as illustrated in Figure 5. (a).

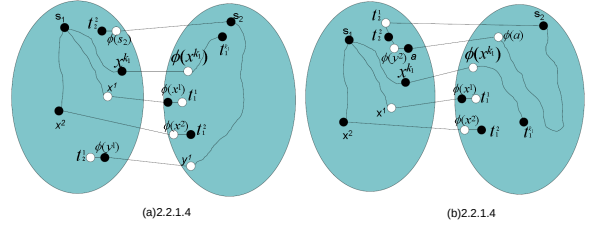


Figure 5: The case 2.2.1.4 of Theorem 1.

Suppose $d(s_2, t_2^2) = 2$ and $d(s_2, t_2^1) = 1$. Let $\phi(x^i, t_1^i) \in Q_n^1$ such that $x^i \notin \{s_1, t_2^1, t_2^2\}$ and $\phi(x^i) \notin (F_a \cup T_1 \cup \{s_2\})$ for $1 \leq i \leq k_1$. Let $(\phi(y^2), t_2^2)$ and $(a, \phi(y^2)) \in Q_n^0$ such that $\phi(y^2), a \notin \{s_1, t_2^1, t_2^2, x^1, \dots, x^{k_1}\}$ and $\phi(a) \notin (F_a \cup T_1 \cup \{\phi(x^1), \dots, \phi(x^{k_1})\})$. Applying Lemma 2, we can construct two spanning disjoint paths $P(s_2, \phi(a)), P(\phi(x^{k_1}), t_1^{k_1})$ of $Q_n^1 - (F_a \cup \{t_1^i, \phi(x^i) | 1 \leq i \leq k_1 - 1\})$. Applying Lemma 4, we can construct a k_1^* -fan $A(s_1 \rightarrow \{x^1, \dots, x^{k_1}\})$ of $Q_n^0 - \{t_2^1, t_2^2, a, \phi(y^2)\}$. Thus, the paths $\langle s_1 \xrightarrow{P(s_1, x^i)} x^i, \phi(x^i), t_1^i \rangle$, $\langle s_1 \xrightarrow{P(s_1, x^{k_1})} x^{k_1}, \phi(x^{k_1}) \xrightarrow{P(\phi(x^{k_1}), t_1^{k_1})} t_1^{k_1} \rangle$, $\langle s_2 \xrightarrow{P(s_2, \phi(a))} \phi(a), a, \phi(y^2), t_2^2 \rangle$ form the $(n-1-f_a)^*$ -bifan of $Q_n - F_a$ for $1 \leq i \leq k_1 - 1$, as illustrated in Figure 5. (b).

Case 2.2.2 $|T^2| = 1$

Case 2.2.2.1 $|F_a^0| = 0$.

We can obtain that $|T_1^0| = 1$. Without loss of generality, we can assume that $T_1^0 = \{t_1^{k_1}\}$. Let $\phi(x^i, t_1^i) \in Q_n^1$ such that $x^i \notin (\{s_1, t_1^{k_1}\})$ and $\phi(x^i) \notin$

$(F_a \cup T_1 \cup \{s_2, t_2^1\})$ for $1 \leq i \leq k_1 - 1$. Suppose $t_2^1 \in Q_n^1$. Applying Lemma 2, we can construct two spanning disjoint paths $P(\phi(x^1), t_1^1)$ and $P(s_2, t_2^1)$ of $Q_n^1 - (F_a \cup \{t_1^i, \phi(x^i) | 2 \leq i \leq k_1 - 1\})$. Applying Lemma 4, we can construct a k_1^* -fan $A(s_1 \rightarrow \{x^1, \dots, x^{k_1-1}, t_1^{k_1}\})$ of Q_n^0 . Thus, the paths $\langle s_1 \xrightarrow{P(s_1, x^1)} x^1, \phi(x^1) \xrightarrow{P(\phi(x^1), t_1^1)} t_1^1 \rangle$, $\langle s_1 \xrightarrow{P(s_1, x^i)} x^i, \phi(x^i), t_1^i \rangle$, $P(s_1, t_1^{k_1})$, $P(s_2, t_2^1)$ form the $(n - 1 - f_a)^*$ -bifan of $Q_n - F_a$ for $2 \leq i \leq k_1 - 1$.

Suppose $t_2^1 \in Q_n^0$. Let $(\phi(y^1), t_2^1) \in Q_n^1$ such that $\phi(y^1) \notin \{s_1, t_1^{k_1}, x^1, \dots, x^{k_1-1}\}$ and $y^1 \notin (F_a \cup T_1 \cup \{\phi(x^1), \dots, \phi(x^{k_1})\})$. Applying Lemma 2, we can construct two spanning disjoint paths $P(\phi(x^1), t_1^1)$ and $P(s_2, y^1)$ of $Q_n^1 - (F_a \cup \{t_1^i, \phi(x^i) | 2 \leq i \leq k_1 - 1\})$. Applying Lemma 4, we can construct a k_1^* -fan $A(s_1 \rightarrow \{x^1, \dots, x^{k_1-1}, t_1^{k_1}\})$ of $Q_n^0 - \{t_2^1, \phi(y^1)\}$. Thus, the paths $\langle s_1 \xrightarrow{P(s_1, x^1)} x^1, \phi(x^1) \xrightarrow{P(\phi(x^1), t_1^1)} t_1^1 \rangle$, $\langle s_1 \xrightarrow{P(s_1, x^i)} x^i, \phi(x^i), t_1^i \rangle$, $P(s_1, t_1^{k_1})$, $\langle s_2 \xrightarrow{P(s_2, y^1)} y^1, \phi(y^1), t_2^1 \rangle$ form the $(n - 1 - f_a)^*$ -bifan of $Q_n - F_a$ for $2 \leq i \leq k_1 - 1$.

Case 2.2.2.2 $|F_a^0| \neq 0$.

Let $(\phi(x^i), t_1^1) \in Q_n^1$ such that $x^i \notin (F_a^0 \cup \{s_1\})$ and $\phi(x^i) \notin (F_a^1 \cup T_1 \cup \{s_2, t_2^1\})$ for $1 \leq i \leq k_1$. Suppose $t_2^1 \in Q_n^1$. Applying Lemma 2, we can construct two spanning disjoint paths $P(\phi(x^1), t_1^1)$ and $P(s_2, t_2^1)$ of $Q_n^1 - (F_a^1 \cup \{t_1^i, \phi(x^i) | 2 \leq i \leq k_1\})$. Applying Lemma 4, we can construct a k_1^* -fan $A(s_1 \rightarrow \{x^1, \dots, x^{k_1}\})$ of $Q_n^0 - F_a^0$. Thus, the paths $\langle s_1 \xrightarrow{P(s_1, x^1)} x^1, \phi(x^1) \xrightarrow{P(\phi(x^1), t_1^1)} t_1^1 \rangle$, $\langle s_1 \xrightarrow{P(s_1, x^i)} x^i, \phi(x^i), t_1^i \rangle$, $P(s_2, t_2^1)$ form the $(n - 1 - f_a)^*$ -bifan of $Q_n - F_a$ for $2 \leq i \leq k_1$.

Suppose $t_2^1 \in Q_n^0$. Let $(\phi(y^1), t_2^1) \in Q_n^1$ such that $\phi(y^1) \notin (F_a^1 \cup \{s_1, x^1, \dots, x^{k_1}\})$ and $y^1 \notin (F_a^1 \cup T_1 \cup \{\phi(x^1), \dots, \phi(x^{k_1})\})$. Applying Lemma 2, we can construct two spanning disjoint paths $P(\phi(x^1), t_1^1)$ and $P(s_2, y^1)$ of $Q_n^1 - (F_a^1 \cup \{t_1^i, \phi(x^i) | 2 \leq i \leq k_1\})$. Applying Lemma 4, we can con-

struct a k_1^* -fan $A(s_1 \rightarrow \{x^1, \dots, x^{k_1}\})$ of $Q_n^0 - F_a^0 - \{t_2^1, \phi(y^1)\}$. Thus, the paths $\langle s_1 \xrightarrow{P(s_1, x^1)} x^1, \phi(x^1) \xrightarrow{P(\phi(x^1), t_1^1)} t_1^1 \rangle$, $\langle s_1 \xrightarrow{P(s_1, x^i)} x^i, \phi(x^i), t_1^i \rangle$, $\langle s_2 \xrightarrow{P(s_2, y^1)} y^1, \phi(y^1), t_2^1 \rangle$ form the $(n - 1 - f_a)^*$ -bifan of $Q_n - F_a$ for $2 \leq i \leq k_1$.

Case 2.3 $|T_1^1| + |F_a^1| = n - 2$.

We can obtain that $(T_1 \cup F_a) \subset Q_n^1$. Let $(\phi(x^i), t_1^1) \in Q_n^1$ such that $x^i \neq s_1$ and $\phi(x^i) \notin (F_a \cup \{t_1^1, t_2^1\})$ for $3 \leq i \leq k_1$.

Suppose $t_2^1 \in Q_n^1$. Applying Lemma 2, we can construct two spanning disjoint paths $\langle t_1^1 \xrightarrow{P(t_1^1, \phi(x^1))} \phi(x^1), \phi(x^2) \xrightarrow{P(\phi(x^2), t_2^1)} t_2^1 \rangle$ and $P(s_2, t_2^1)$ of $Q_n^1 - (F_a \cup \{t_1^i, \phi(x^i) | 3 \leq i \leq k_1\})$. Applying Lemma 4, we can construct a k_1^* -fan $A(s_1 \rightarrow \{x^1, \dots, x^{k_1}\})$ of Q_n^0 . Thus, the paths $\langle s_1 \xrightarrow{P(s_1, x^1)} x^1, \phi(x^1) \xrightarrow{P(\phi(x^1), t_1^1)} t_1^1 \rangle$, $\langle s_1 \xrightarrow{P(s_1, x^2)} x^2, \phi(x^2) \xrightarrow{P(\phi(x^2), t_2^1)} t_2^1 \rangle$, $\langle s_1 \xrightarrow{P(s_1, x^i)} x^i, \phi(x^i), t_1^i \rangle$, $P(s_2, t_2^1)$ form the $(n - 1 - f_a)^*$ -bifan of $Q_n - F_a$ for $3 \leq i \leq k_1$, as illustrated in Figure 6. (a).

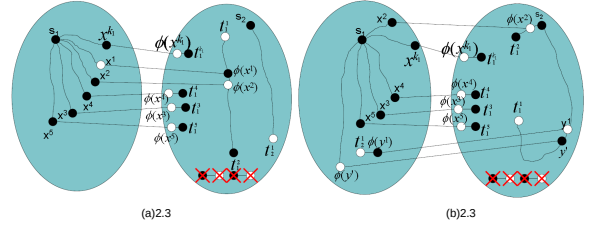


Figure 6: The case 2.3 of Theorem 1.

Suppose $t_2^1 \in Q_n^0$. Applying Lemma 1, we can construct a Hamiltonian path $P(t_1^1, t_2^1)$ of $Q_n^1 - (F_a \cup \{t_1^i, \phi(x^i) | 3 \leq i \leq k_1\})$. Let $(\phi(y^1), t_2^1) \in Q_n^0$ where y^1 is on the path $P(t_1^1, t_2^1)$ such that (y^1, y') is on the path $P(t_1^1, t_2^1)$ but not on the subpath between y^1 and s_2 for $\phi(y') \neq t_2^1$ and $\phi(y^1) \notin (\{s_1, x^3, \dots, x^{k_1}\})$. Without loss of generality, we can denote this path as $\langle t_1^1 \xrightarrow{P(t_1^1, y')} y', y^1 \xrightarrow{P(y^1, s_2)} s_2, \phi(x^2) \xrightarrow{P(\phi(x^2), t_2^1)} t_2^1 \rangle$. Applying Lemma 4, we can construct a k_1^* -fan $A(s_1 \rightarrow \{y', x^2, \dots, x^{k_1}\})$ of $Q_n^0 - (\{t_2^1, \phi(y^1)\})$. Thus, the paths $\langle s_1 \xrightarrow{P(s_1, \phi(y'))} \phi(y'), y' \xrightarrow{P(y', t_1^1)} t_1^1 \rangle$, $\langle s_1 \xrightarrow{P(s_1, x^2)} x^2, \phi(x^2) \xrightarrow{P(\phi(x^2), t_2^1)} t_2^1 \rangle$, $\langle s_1 \xrightarrow{P(s_1, x^i)} x^i, \phi(x^i), t_1^i \rangle$, $\langle s_2 \xrightarrow{P(s_2, y^1)} y^1, \phi(y^1), t_2^1 \rangle$ form the $(n - 1 - f_a)^*$ -bifan of $Q_n - F_a$ for $3 \leq i \leq k_1$.

$y^1, \phi(y^1), t_2^1$ form the $(n - 1 - f_a)^*$ -bifan of $Q_n - F_a$ for $3 \leq i \leq k_1$, as illustrated in Figure 6. (b).

4 Conclusion

In this paper, we have shown that $Q_n - F_a$ is $(n - 1 - f_a)^*$ -bifanable with the same color sources if $f_a \leq n - 3$ where F_a is the set of f_a pairs of adjacently faulty vertices. The vertices fault-tolerance and edges fault-tolerance for fanability and bifanability are worth studying in the future.

References

- [1] Y. H. Chang, C. N. Hung, "Adjacent Vertices Fault-tolerance Hamiltonian Laceability of Hypercube," *Workshop on Combinatorial Mathematics and Computational Theory*, 22 (2005), pp.301-309.
- [2] C. H. Chang, C. K. Lin, H. M. Huang, and L. H. Hsu, "The super laceability of the hypercubes," *Information Processing Letters*, Vol. 92, pp. 15-21, 2004.
- [3] C.C. Chen, C.N. Hung, K.C. Hu, "Edge Fault-tolerant of k^* -bifanability for bipartite Hypercube-like graphs," *Workshop on Combinatorial Mathematics and Computational Theory*, 22 (2005), pp.134-139.
- [4] C. Hung and P. Lin, "The Study for Adjacent Vertices Fault-Tolerance Bifanability of Hypercube," *Proceedings of the 2009 National Computer Symposium Workshop on Algorithms and Bioinformatics*, 2009, pp. 215-224.
- [5] C.D. Park, K.Y. Chwa, "Hamiltonian properties on the class of hypercube-like network," *Information Processing Letters*, 91 (2004), pp.11-17.
- [6] W.Y. Su, C.N. Hung, "The longest ring embedding in faulty hypercube," *Workshop on Combinatorial Mathematics and Computational Theory*, 23 (2006), pp.262-272
- [7] C. H. Tsai, J.J.M. Tan, T.Liang, and L.H. Hsu, "Fault-tolerant Hamiltonian laceability of hypercubes," *Information Processing Letters*, 83, pp.301-306, (2002).